

Polycystic ovarian syndrome

Presented by:

Dr. ROZHAN YASSIN KHALIL

Assistant professor & consultant obstetrics & gynecology
FICOG, CABOG, HDOG, FICS, MBChB

2025

Objective:

- Understand the epidemiology and effects of polycystic ovary syndrome (PCOS), its diagnosis and management.
- Causes and risk factors.
- Does life style affect pcos.
- What are the consequence if not treated.

Polycystic ovarian syndrome

Polycystic ovary syndrome PCOS is a **syndrome of ovarian dysfunction along with the cardinal features of hyperandrogenim and polycysticovary morphology.**

The definition according to meeting held in **Rotterdam** may 2003:

The diagnosis of PCOS is made when **two out of three** of the following criteria are found:



even in the presence of
**androgen excess
after exclusion of
other related
disorder.**

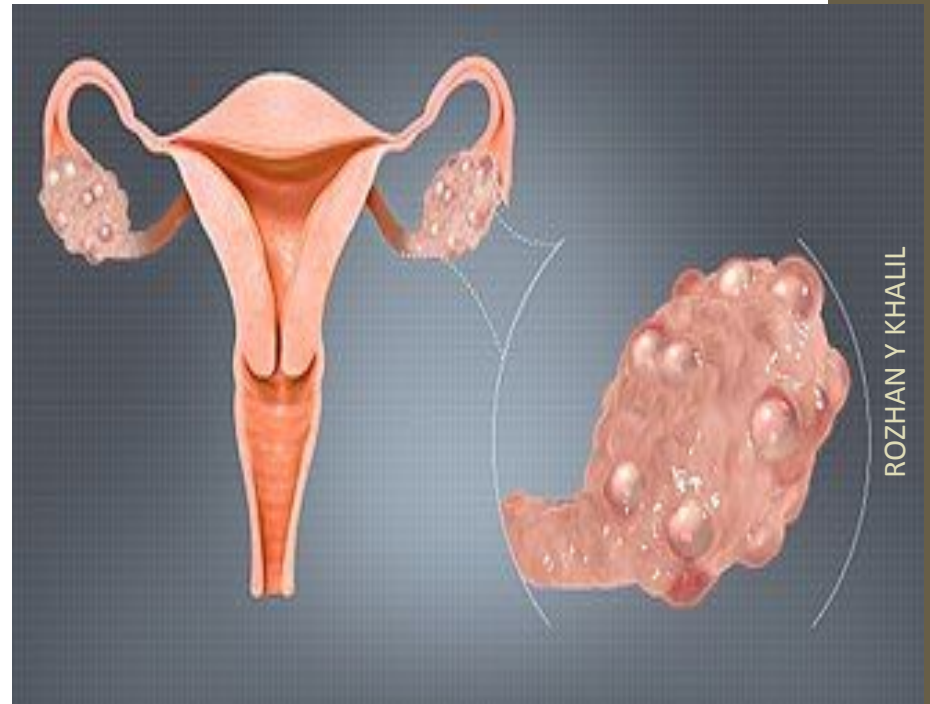


**2. Oligo and
/or
anovulation**



**3. Ultrasound
appearance of the
ovaries with evidence
of >12 follicles in
each ovary
measuring 2-9
mm and /or
increased ovarian
volume (>10 ml).**







Prevalence

PCOS is thought to occur in about **6-8** per cent of women worldwide, making it the most common reproductive disorder.

The prevalence is also higher in certain **ethnic groups, such as South Asians**, who may also suffer from more severe symptoms.

AETIOLOGY

The aetiology of PCOS is largely unknown ,but seems to involve a complex interaction between **enviromental (e.g. diet and exercise)and multiple genetic factors.**

The mode of inheritance appears
(**autosomal inheritance**).

Several factors have been implicated in the pathogenesis of PCOS, including



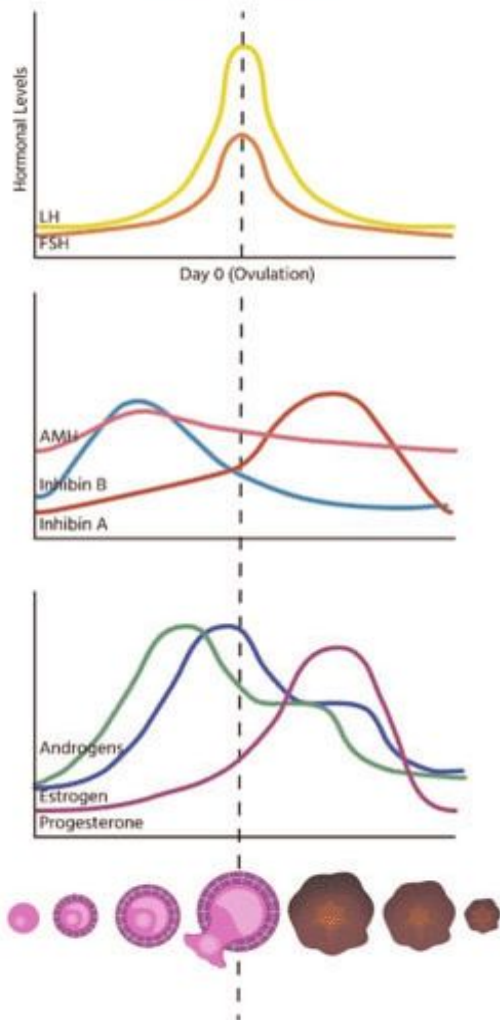
Pathophysiology:

1.a dysfunction of ovarian function characterized by increased production of ovarian androgens.

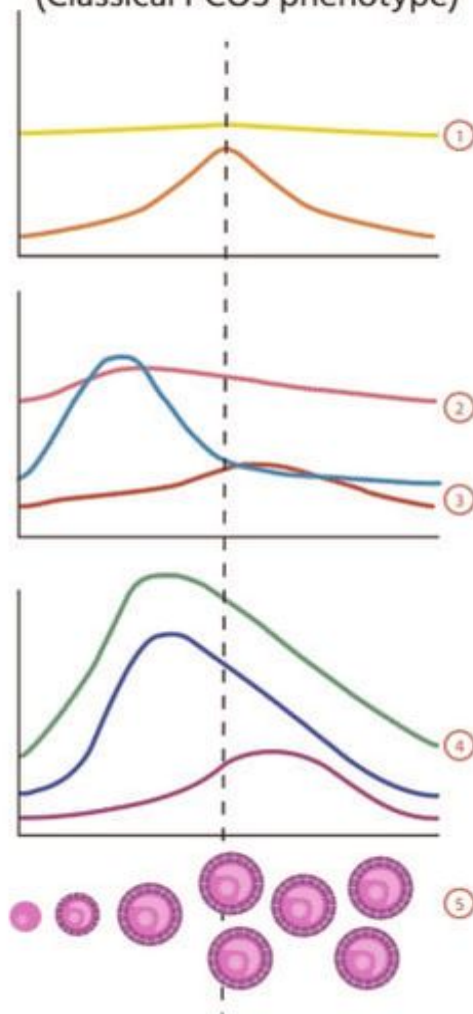
2.a dysfunction in hypothalamic function resulting in increasing **LH** which stimulates androgen production by theca cells.

3.insulin resistance which is characteristic of both obese and non obese pcos.

A Normal Menstrual Cycle



B PCOS Comparison (Classical PCOS phenotype)



C Explanation of Differences in PCOS females

- 1 **LH increases** across the cycle due to inconsistent GnRH pulsatility in females with PCOS. LH goes on to stimulate androgen production (4) and influence folliculogenesis (5).
- 2 **AMH is elevated** in females with PCOS. This is thought to reflect the increased antral follicle count (5) since antral follicles release AMH to inhibit the recruitment of further primordial follicles.
- 3 **Inhibin B is also elevated** in females with PCOS. Inhibin B is also secreted by antral follicles to suppress primordial follicle recruitment. Since no dominant follicle is selected, Inhibin A levels, secreted by dominant follicle, are negligible.
- 4 **Androgen levels are elevated across the cycle** in females with PCOS. Elevated androgen levels can contribute to follicular arrest (5) and acyclical estradiol conversion, which leads to inconsistent negative feedback on GnRH pulsatility, affecting LH secretion (1).
- 5 **There is an increased number of antral follicles** in females with PCOS. This results from premature follicular arrest, which creates anovulatory cycles as well as increased AMH (2) and inhibin B secretion by the antral follicles (3).

References: [7,23,29,80]

Clinical features

Oligomenorrhoea/
amenorrhoea in up to
75% of patients.

Subfertility in up to 75%
of women.

Clinical features:

dermatological manifestaions are the reflection of the metabolic and biochemical disturbances. The most common of these is hirsutism, other changes include oily skin,seborrhoea ,alopecia ,acne and acanthosis nigricans.

Hirsutism ,acne ,alopecia and oily skin are all manifestations of hyperandrogenism,

on the other hand ,acanthosis nigricans ,which is characterized by pigmented ,velvety skin patches in the axilla,nape of the neck, antecubital fossa and groin,is a marker of insulin resistance.

A



B



OTHER CLINICAL MANIFESTATIONS OF PCOS:

- OBESITY: obesity 50 per cent or more of patients with PCOS are obese.
- The metabolic syndrome; Women suffering from insulin resistance are at increased risk of developing the **metabolic syndrome characterized by: type 2 diabetes, hypertension, dyslipidaemia, atherosclerosis and ischemic heart disease.**

Management

Even if normalization of ovulation is not required, then it is still important to protect the endometrium against the long term effects of prolonged **unopposed oestrogen** using, for example **progesterone or the oral contraceptive pills.**

The treatment of PCOS depends on whether or not fertility is desired.

If fertility were required then the aim would be to induce ovulation using oral anti-oestrogen, gonadotrophins or more recently aromatase inhibitors. Insulin sensitizing agents are also commonly used to reduce insulin resistance and consequent hyperandrogenism,

TREATMENT OF OBESITY

LIFESTYLE CHANGES

In obese PCOS patients ,weight loss is the cornerstone of treatment , when even a small amount of weight loss can lead to spontaneous resumption of ovulation.

Weight loss is also important prior to conception to minimize the associated risks of maternal and fetal morbidities.

Weight loss is best achieved through combination of diet and exercise.

PHARAMACOLOGICAL

Some obese PCOS however, will find it difficult to achieve adequate weight loss with diet and exercise alone in which case the weight-losing drugs may be indicated

There are two weight losing drugs currently licensed:

The centrally acting serotonin and norepinephrine uptake inhibitor **(sibutramine)**

the peripheral acting lipase inhibitor **(orlistat).**

Both medications have been shown to be effective at producing a modest weight loss together with a well balanced diet

However in PCOS patients, it needs to be remembered that fertility treatment is often the main target and therefore the safety of these drugs in the presence of an early pregnancy needs to be considered

BARIATRIC SURGERY

The guidelines of the NATIONAL INSTITUTE for HEALTH and CLINICAL EXCELLNCE (NICE) state that

,

surgery is recommended as a treatment option for people with morbid obesity (body mass index (BMI) equal to or greater than 40 kg /m²

or with BMI equal to or greater than 35 kg/m² in the presence of significant co-morbid conditions that could be improved by weight loss

OVULATION INDUCTION

ANTI-OESTROGENS

Clomifene citrate has been used for ovulation induction in women with anovulatory infertility for over 40 years.

Clomifene citrate can be offered for 6-12 months provided there is evidence of ovulation.

The main side effect is that of multiple pregnancies, which may occur in **6-8 per cent** of cases, tamoxifen is another anti-oestrogen similar to clomifene.

OVULATION INDUCTION

Gonadotropins

FSH is mainly indicated in cases of clomifene resistance and in those who fail to conceive or have intolerable side effects with clomifene.

- associated with an increased risk of multiple follicular development and multiple pregnancies.
- PCOS patients are particularly prone to ovarian hyperstimulation syndrome(OHSS).
- .

OVULATION INDUCTION

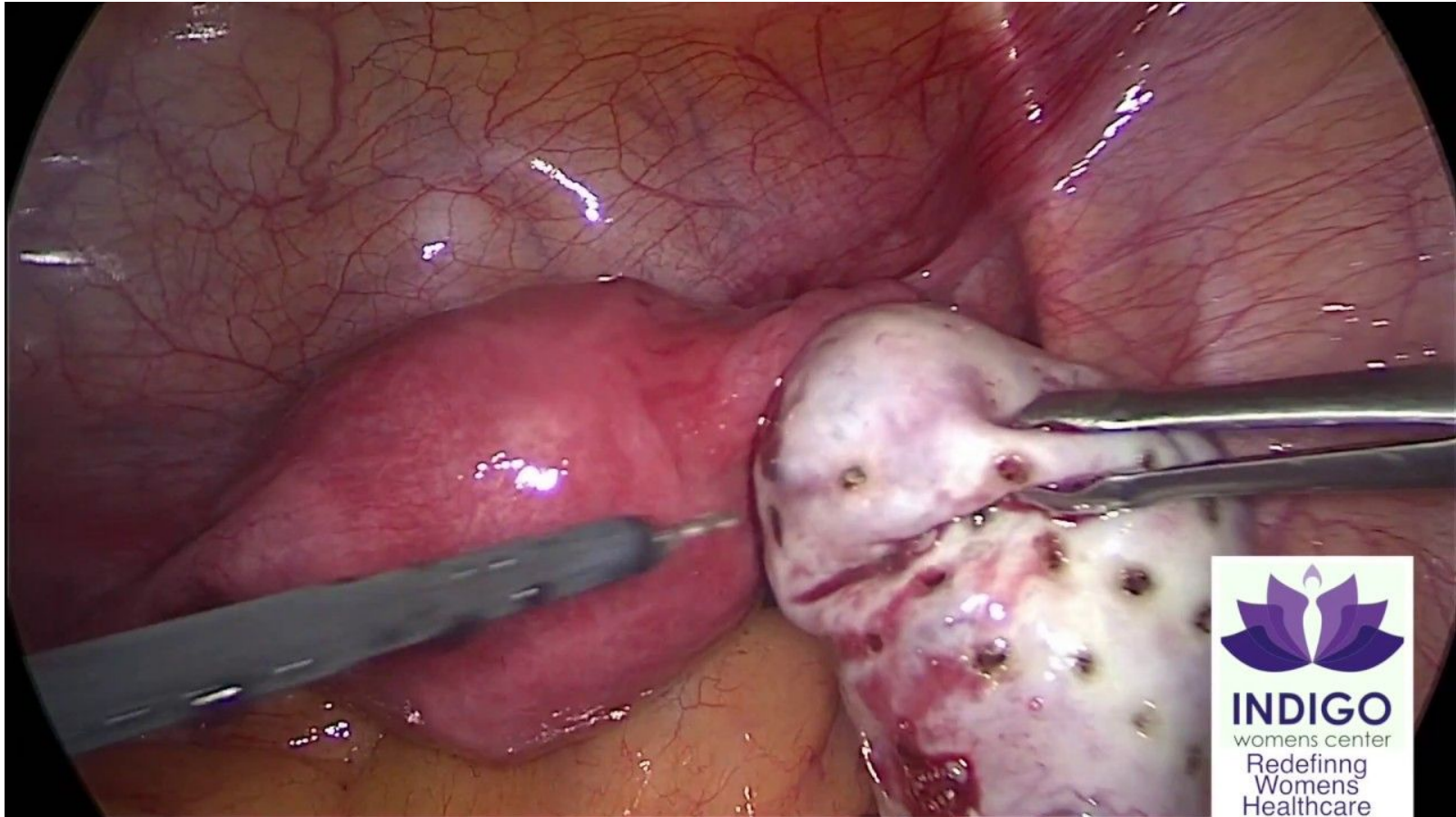
GnRH agonists :Its main use is in women with hypogonadotropic hypogonadism;however,in women with PCOS,there is no clear evidence that GnRH treatment is associated with better pregnancy rates compared to gonadotropins alone or in combination

Aromatase inhibitor :3rd generation aromatase inhibitor (anastrozole and letrozole), these drugs inhibit ovarian aromatase activity leading to a decrease in circulating oestradiol concentration and consequent increase in FSH release.

Laparoscopic ovarian drilling

Is mainly indicated in clomifene –resistant patients where it has been shown to have a similar efficacy to gonadotropins with the advantage of a lower multiple pregnancy rate.

LOD can result in ovulation in approximately 80 per cent and a clinical pregnancy in 60 per cent of cases.



TREATMENT OF HIRSUTISM

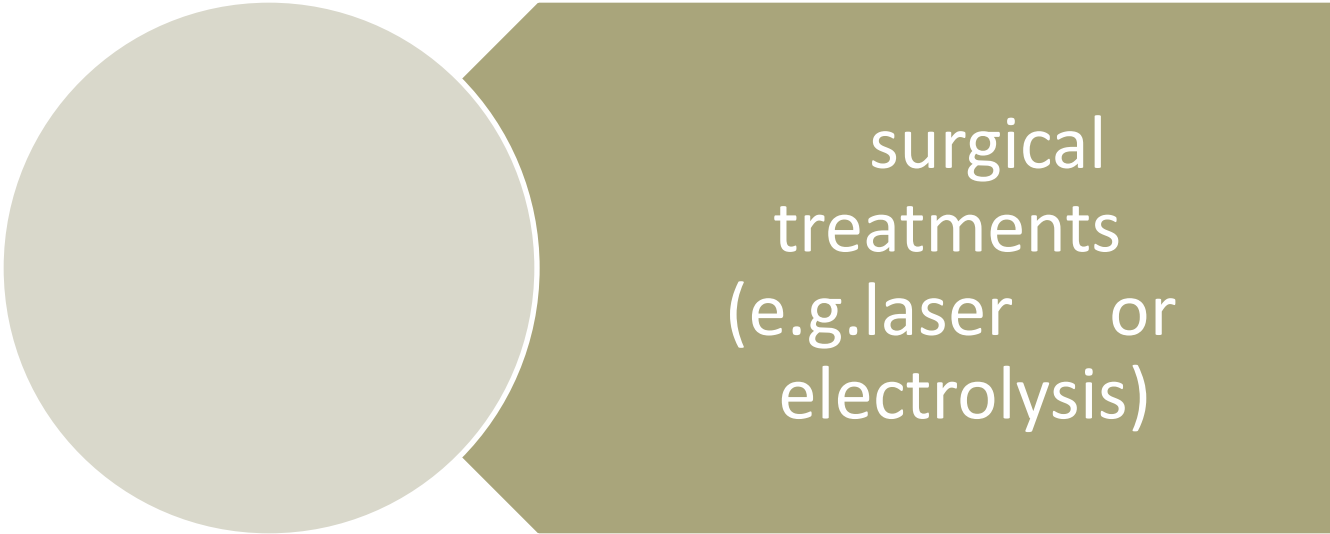
ORAL CONTRACEPTIVE PILL
:The OCP is usually the first line of therapy ,particularly for those requiring contraception or for those with period irregularities

ANDROGEN ANTAGONISTS :These medication are usually used as second-line monotherapy or in combinaton with the OCP in severe cases.(Spironolactone,Flutamide ,Finsteride).

Eflornithine(vaniqa) is a topical drug that act locally, it can result in visible improvement within a few weeks

INSULIN SENSITIZING AGENTS :Metformin may improve hirsutism through an improvement in insulin resistant

GnRH AGONISTS :Can be used to suppress pituitary gonadotropins and ovarian activity in severe resistant cases,however treatment is associated with significant menopause.

A graphic consisting of a light gray circle on the left and a dark olive green hexagon on the right, which overlaps the circle. The text is centered within the hexagon.

surgical
treatments
(e.g.laser or
electrolysis)

Conclusions:

- Pcos is regarded as chronic disease.
- Life style change is corner stone in the treatment.
- Long term morbid consequence can result if un treated.



THANKS